

Lumbar Spine Treatment

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A simple approach to start the treatment plane

- Use the subjective and objective assessment
- Clear Red flags!
- Clinically diagnose what the condition is using different approaches such as pain source, movement response
- Formulate a problem list
- Prepare a treatment plane to resolve the list of problems.
- Treatment plan should include the highest evidence-based practice:

    1- education

    2- Exercises

    3- might be able to use adjuncts such as manual therapy.

What does best practice care for musculoskeletal pain look like? Eleven consistent recommendations from high-quality clinical practice guidelines: systematic review

1. Care should be patient centred.
2. Screen patients to identify those with a higher likelihood of serious pathology/red flag conditions.
3. Assess psychosocial factors.
4. Radiological imaging is discouraged unless:
    1. Serious pathology is suspected.
    2. There has been an unsatisfactory response to conservative care or unexplained progression of signs and symptoms.
    3. It is likely to change management.
5. Undertake a physical examination, which could include neurological screening tests, assessment of mobility and/or muscle strength.
6. Patient progress should be evaluated including the use of outcome measures.
7. Provide patients with education/information about their condition and management options.
8. Provide management addressing physical activity and/or exercise.
9. Apply manual therapy only as an adjunct to other evidence-based treatments.
10. Unless specifically indicated (e.g. red flag condition), offer evidence-informed non-surgical care prior to surgery.
11. Facilitate continuation or resumption of work.

Problems in Lumbar spine conditions

1- Pain such as pain during movement or radicular pain: pain management, ROM exercises (preferred directions), neurodynamic mobility, manual therapy and other adjuncts.

2- Movement / mobility/ Balance problems as reduced lumbar range or compensatory movement (thorax, hip), asymmetrical movement (motor control).

3- Strength / motor control problems as reduced (delayed) deep core activation, weak hip muscles (altered lumbopelvic control), reduced endurance of trunk stabilisers, or overactivation of(thoracic or lumbar or hip) muscles

Problems in Lumbar spine conditions

4- Functional problems: walking, sitting, standing, squatting, picking up objects.

5- Postural problems: hyperlordosis, sway back, lateral shift, lower crossed syndrome, S scoliosis

6- Education or Knowledge gap or Psychological /behavioural problems: fear avoidance, reduced confidence in capabilities, anxiety about re-injury limiting participation in exercise, limited understanding or misunderstanding of pain mechanisms or causes or pain chronicity, poor awareness of posture, how to modify activities to reduce symptoms, how to exercises safely during flare-ups

Pain Management

- Physiotherapy is commonly a key aspect in managing chronic pain, being part of this multidisciplinary, bio-psycho-social approach.
- Evidence has shown that physiotherapists use a variety of skills to guide and support people with chronic pain towards a better quality of life.
- Despite progress in chronic pain management, some patients and some clinicians still expect a ‘cure’. It is often assumed that ‘hands-on’ physiotherapy will be offered, (which is commonly not indicated or appropriate).
- Certain patients fear physiotherapy, exercise and activity, as past experience has worsened their pain or not helped at all.

Pain Management EBP

Intervention	Description
1. Education	General education on injury/disease, acute and chronic pain-physiology, neuroscience education, regimes, activity-modification, ergonomics, general health, coping strategies
2. Sensorimotor training	Can also be termed neuromuscular training or motor control exercises and refers to exercises with the purpose of improving sensorimotor control - including training of muscular synergies, balanced in time and magnitude through the relearning of motor control - in specific, the exercises aim at creating appropriate, automatic and generalized movements to optimize muscular joint stabilization commonly affected by the pain
3. Physical activity-advice	Individualized advice on how to optimize the individual patient's physical activity despite the chronic pain, most often advice on walking, Nordic walking, cross-training or cycling.
4. Interventions aiming at improving structures and functions	Stretching, manual physical therapy techniques, heat or cold treatment, spray-n-stretch-technique, taping – all prior to exercises for movement or stabilization
5. Sensory stimulation	Transcutaneous electrical nerve stimulation, acupuncture
6. Physical activity carried out together with the physiotherapist	Physical activity with direct guidance on for example exercise-bike, cross-trainer or Nordic walking
7. Weight training	Individualized exercises performed in a gym center or weight training with other equipment
8. Relaxation, Mindfulness	Relaxation techniques such as breathing exercises, progressive muscle relaxation, autogenous training. Mindfulness-training including paying attention to thoughts, feelings, bodily sensations other than pain in the present moment, often while carrying out movements
9. Physiotherapy interventions with an Acceptance and commitment therapy (ACT)-inspired approach	Interventions to support acceptance, committed action, values, movement- and behavior change and to reduce fear-of-movement, and also specific interventions on motivation - MI, motivating interviewing - aiming at improving self-efficacy in behavior-change
10. Basic body awareness therapy and training of specified activities	A movement-based physiotherapeutic method developed in Scandinavia where movements aim at enhance body awareness and consciousness of the body with the purpose to move with less effort

Analgesic adjuncts & referral for medication review (collaborative care)

- Coordinating care and referring for medication review (NSAIDs, neuropathic agents) is often appropriate.
- Appropriate medication can enable engagement with active rehab when pain is limiting; always weigh benefits vs harms.
- Discuss with GP/pain clinic when patient's pain prevents participation, or when red flags/complex pharmacology exist.
- Follow guidance in NICE NG193 for chronic pain pathways. NICE



Exercise prescription for the Lumbar spinal conditions



Therapeutic Exercise

Evidence from systematic reviews support the use of therapeutic exercise to improve MSK pain and chronic illnesses in several conditions.

Incorporation of exercise and PA and their importance are usually introduced during initial individual appointments and continued through to structured exercise classes or as a component of a chronic pain rehabilitation program.

Using frequency, intensity, type, and time (duration) parameters (F.I.T.T.) for specific exercises.

The choice of exercise therapy should be closely aligned with patients' preferences and goals as enjoyment in and commitment to the type of exercise will help with long-term adherence

Types of Exercises

1- Strengthening exercise: Endurance (Muscle), strength and power

2- Stretching Exercises

3- Balance Exercises

4- Aerobic Exercises

5- Range of motion Exercises

6- Functional Exercises

7- Perceptual/ Motor Control Exercises



Application of therapeutic exercises

Principles of exercise prescription: last year's teaching

Balance problems: review balance rehabilitation from last year.

Specific exercises for core muscles: review the MMT in the lumbar spine assessment Stretching, flexibility and PNF: concepts last year's teaching.

Aerobic Exercises: concepts last year's teaching

Range of motion Exercises: similar to ROM assessment (lumbar assessment), choose between active, passive, and active assisted.

You can also identify the needed exercises by connecting exercises and function

Functional task	Movement/exercise required
Moving from sitting to standing	Forward flexion of the lumbar spine
Increased standing and walking tolerance	Contracting the quadriceps and gluteal muscle groups
Getting out of bed	Pelvic tilts to mobilize the lumbar spine
Walking to the toilet	Exercising the hamstring and quadriceps muscle groups/balance exercises

McKenzie Method (Mechanical Diagnosis and Therapy - MDT)

- Involves repeated movements or sustained postures to centralize or reduce spinal symptoms.
- Often used for discogenic pain, sciatica, and chronic spinal conditions. decreases pain centralises leg symptoms toward the spine increases movement quality or improves function
- Long et al. (2004) demonstrated better outcomes with MDT in patients with directional preference compared to standard physiotherapy.
- A systematic review by May et al. (2018) found McKenzie therapy superior to passive treatments for lumbar radiculopathy.

Mackenzie Method



Centralization is the symptoms movement from the distal segments of body to the proximal. Advent of centralization is a good signal and speaks of a directional preference.



And in contrast, Peripheralization - the movement of pain from the spine to the extremities, indicates a worsening.



Mackenzie technique is applied to patients who have referred pain from the spine to the extremities.

McKenzie Method (Mechanical Diagnosis and Therapy - MDT)

A. Patients perform repeated movements uses repeated movements in the patient's preferred direction — the direction that reduces, centralises, or improves symptoms. (lumbar extensions for posterior disc bulge).
    - Repeated flexion in lying (knees-to-chest), or in sitting or in standing

B. Sustained positions that brings comfort:
    - sustained prone lying or standing in extension
    - sustained flexion in sitting (rare, only when clinically appropriate)
    - sustained lateral shift correction with manual support

Identify the preferred direction

- Patient feels comfortable in standing, prone lying → Extension exercise
- Patient feels comfortable in sitting, knee to chest → Flexion exercise
- Patient feels comfortable in lateral glide, side shift→ lateral flexion exercise

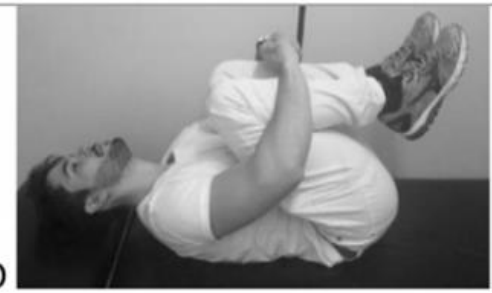
Treatment choice

- If repeated movement helps → prescribe repeated movement.
- If repeated movement is too painful → start with sustained positions.
- If sustained positions help → progress to repeated movement later.

When to stop using this technique

A movement is NOT the preferred direction if it…

1. Increases pain
2. Causes pain to spread further down the leg (peripheralisation)
3. Increases stiffness or guarding
4. Makes movement harder
5. Feels “wrong,” “worse,” “compressed,” or “tightening”

Extension Principle	Flexural Principle	Correction of lateral deviations
A 	D 	G  *
B 	E 	H  **
C 	F 	I  **



Lateral shift correction

- Passive shift correction is often the starting point. The therapist stands on the side of the trunk deviation and applies a sustained translatory force to the thorax in the opposite direction of the shift, while stabilising the pelvis. This is held for several seconds and repeated, observing symptom response. A positive response is reduction in leg pain and gradual realignment of the trunk.
- Once some correction is achieved, active shift correction is encouraged. The patient is taught to move their trunk toward the corrected position while maintaining upright posture, repeating this throughout the day. This promotes self-management and prevents recurrence.
- After the shift begins to reduce, extension-based exercises are usually introduced, such as repeated lumbar extension in standing or lying. These help promote disc material migration anteriorly and reduce posterior nerve root compression.

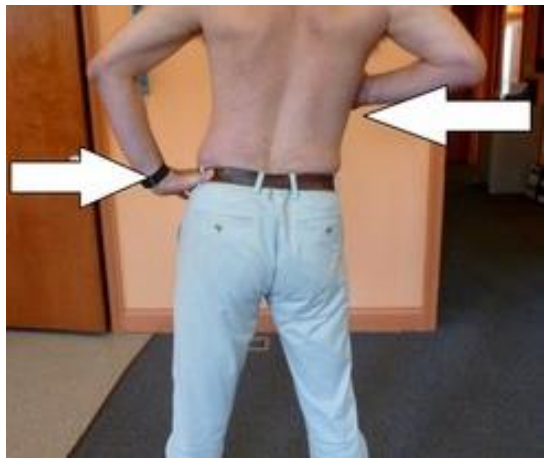


Lateral shift correction

- When passive shift correction is too painful, this usually indicates a highly sensitive nerve root and acute inflammatory phase of disc pathology. Forcing the correction at this stage can lead to peripheralisation of symptoms, increased muscle spasm and further protective guarding, which is counterproductive. The priority then shifts from mechanical realignment to symptom modulation and gradual desensitisation.
- The first step is to find a position of comfort rather than a position of correction. This involves placing the patient in a posture that reduces leg pain, even if the shift remains visible. Often this is side-lying on the non-painful side with pillows supporting the trunk, or lying prone if tolerated. The aim is to reduce neural irritation and allow muscle guarding to settle.
- Instead of direct manual correction, repeated movements within the patient's tolerance are introduced. These are usually very small-range movements toward the direction that centralises symptoms. For example, gentle side-gliding in standing or very mild lateral shifts performed actively by the patient, staying well within pain limits. The key principle is symptom-guided movement, not structural force.

Self Correction of lateral shift









Correction of lumbar lordosis

Improving hip flexor length,

Strengthening gluteals and lower abdominals,

Retraining deep stabilisers,

Enhancing thoracic mobility,

And improving functional movement control.

Patient Education

- Reassurance, and role of emotional factors role in symptoms.
- The aetiology of the pain experience/ their symptoms.
- Pain and pain mechanisms (central sensitization).
- Safety netting
- Promotion of self-management, Self-pacing in chronic cases.
- Value of adherence to treatment plan, Tools to promote long-term adherence (physiotools and pamphlets).
- Correct faulty conceptions (behavioure modifications)
- How to deal with a daily activities and workload
- Activity modification
- Ergonomic adjustments to perform the limited tasks

Education and self-management programs

- Structured self-management interventions including exercise classes, education sessions, and tools to promote long-term adherence.
- Promote independence, reduce healthcare use, and improve function.
- Group formats can be efficient and effective for many patients.
  Provide clear written/digital plans, set SMART goals, schedule follow-ups, and use behavioural strategies to support adherence (habit formation, reminders).
- NICE/CSP guidance and large trials show benefit of structured exercise/education programs for OA and chronic MSK conditions; recent large-scale programs show promising population effects. NICE

Activity pacing and graded return to function

- Systematic, progressive reintroduction of feared or avoided activities (breaking tasks into smaller steps and increasing challenge gradually). Pacing is structuring activity to avoid boom-and-bust patterns.

- Reduces fear-avoidance, builds tolerance, restores function and confidence especially important where kinesiophobia or fear of reinjury exists.

- Identify feared activities, set SMART graded goals, schedule incremental exposure, and combine with PNE and reassurance. Use objective measures to track progress.

- Graded exposure and pacing are core recommendations in biopsychosocial management of chronic pain and are supported by clinical trials for fear-based impairments. NICE supports active rehab and graded approaches in chronic pain pathways. NICE

Neurodynamic Techniques for Neurological symptoms and radiculopathy

- Neural mobilization generally consists of techniques termed neural glides or neural flossing. Neural stretching is also performed as a mobilisation technique. Although glides or flossing is also seen by some as a stretch, they are generally more dynamic in nature.
- Administration of neurodynamic techniques should theoretically restore the dynamic balance between the nerve and surrounding tissues, which in turn should improve the anatomic relations and the physiological function of the nerve.

Neurodynamic Techniques

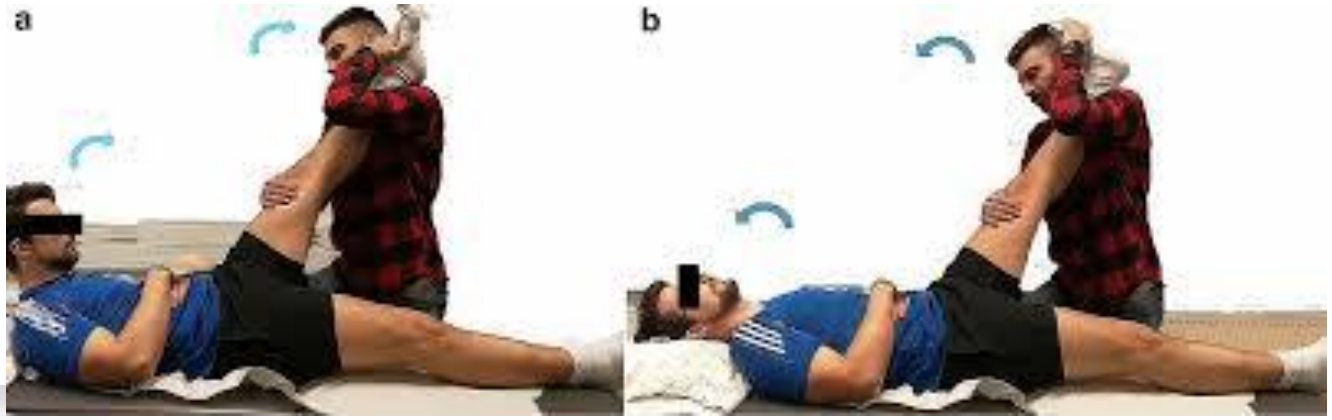
Tensioners

- **Tensioners:** Progressive stretching of neural structures (e.g., **straight leg raise with dorsiflexion** for sciatic nerve glide).

Sliders / Gliders

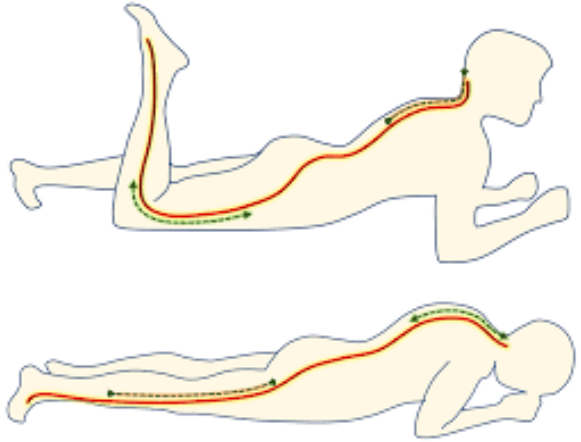
- **Sliders:** Alternating movements that **mobilize the nerve without increasing tension**. Example: Extending the knee while simultaneously extending the cervical spine in sciatic nerve mobilization.

Neurodynamic Techniques



Hypothetical advantages of applying the neurodynamic techniques include reduction of edema and adhesion, improvement of blood supply and nerve nutrition, improvement of axonal transport and nerve conduction, and reduction of pain associated with ischemia.





SLR exercises

NDexer 1:

Start position



Spine neutral, right hip ~30-40° flexion, right knee ~60° flexion

Movement 1



Ankle plantar flexion / Neck flexion

Movement 2



Ankle dorsiflexion / Neck extension

NDexer 2:

Start position



Spine neutral, right hip ~30-40° flexion, right knee fully extended

Movement 1



Ankle plantar flexion / Neck flexion

Movement 2



Ankle dorsiflexion / Neck extension

NDexer 3:

Start position



Spine neutral, right hip ~30-40° flexion, right knee ~60° flexion

Movement 1



Knee flexion / Neck flexion

Movement 2



Knee extension / Neck extension

SLR exercises

NDexer 4:



NDexer 5:



Femoral Nerve



(A) (B)

Manual Therapy for Lumbar spine

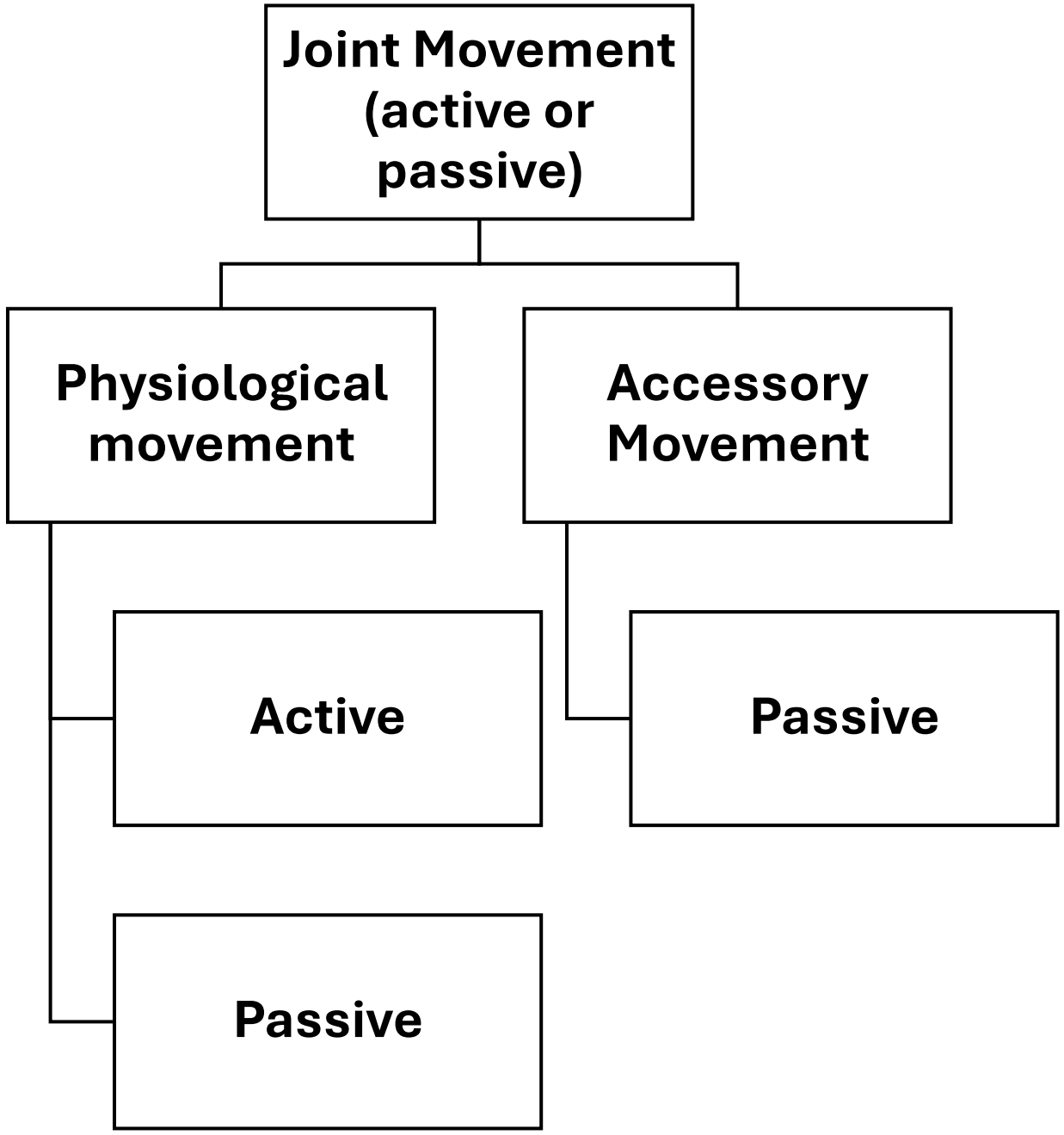
Spinal Manual Therapy

Spinal manual therapy techniques are hands-on interventions used by physiotherapists to assess and treat musculoskeletal dysfunctions of the spine. These techniques aim to reduce pain, improve mobility, and enhance function.

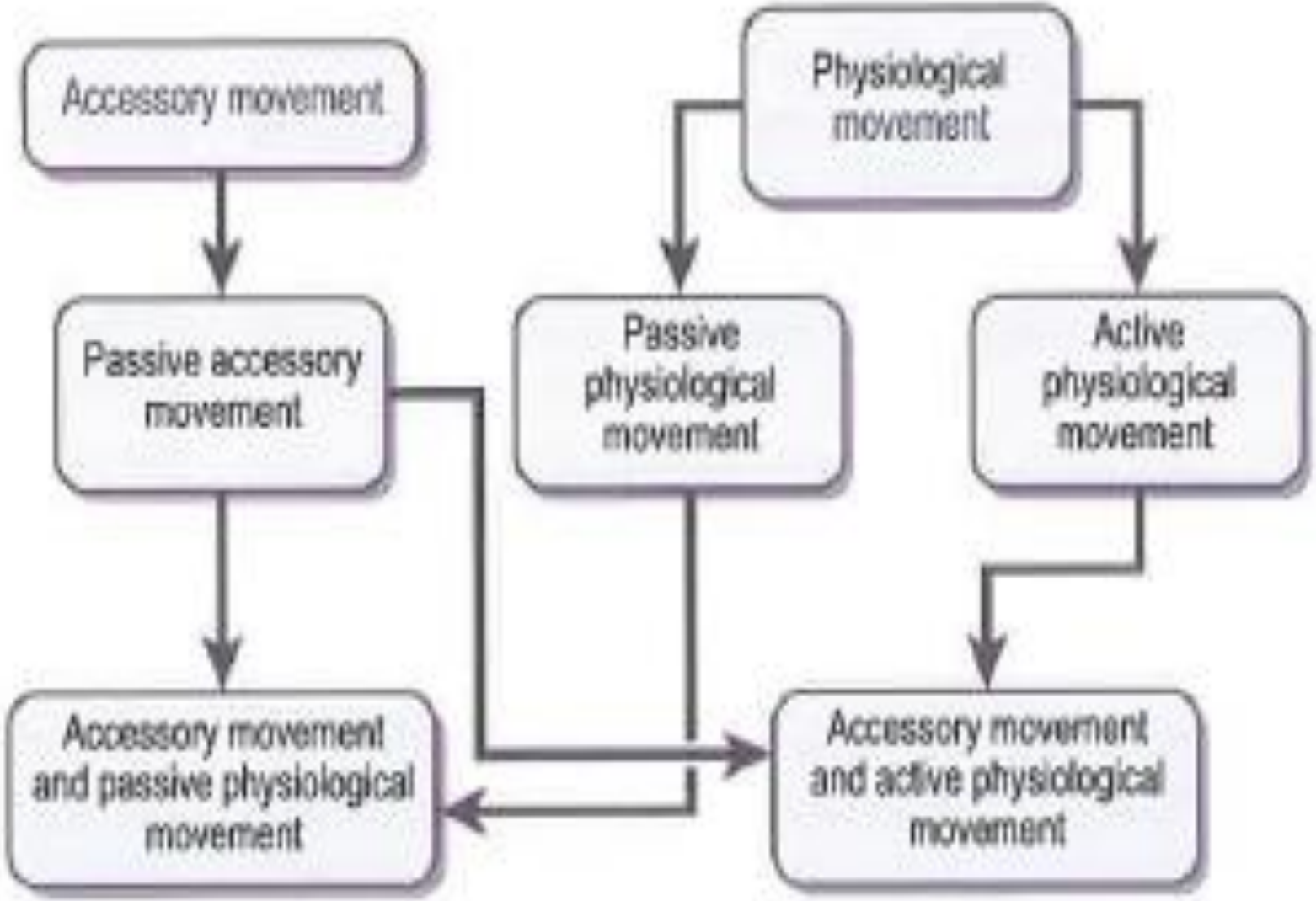
Examples of the MT schools:

- Maitland Technique
- Mulligan Technique
- Myofascial release and Trigger point therapy
- Positional release/ Strain-counter strain technique

Types of Maitland Mobilisation



Types of Maitland Mobilisation



Spinal Maitland Technique

Passive Physiological Intervertebral Movements (PPAIVMs)

Passive Accessory Intervertebral Movements (PAIVMs)

Contraindications to Vertebral Mobilisations

1. Neurological: Disturbance of bladder and bowel function, or perineal anaesthesia, Spinal cord symptoms (e.g. CES or cervical myelopathy).
2. Spinal instability: Down syndrome, Rheumatoid Arthritis and Osteoporosis, Spondylolysis, spondylolisthesis
3. Any pathology leading to significant bone-weakening such as tumours, infections, long-term corticosteroid medication, fracture
4. Patients with vertigo need close supervision
5. Laxity of ligaments: Hypermobility, Pregnancy
6. Aortic aneurysm, bleeding into joints, e.g. severe Haemophilia
7. Musculoskeletal deformity

Maitland Concept Application:

- Grade I & II: within the pain-free range, applied for pain relief.
- Grade III & IV: applied to increase mobility.
- Evidence:
  - A systematic review by Verma et al. (2021) found Maitland mobilizations effective in reducing pain and improving range of motion (ROM) in chronic LBP.
  - Studies by Mintken et al. (2018) showed cervical mobilizations are effective for reducing mechanical neck pain and improving function.

PPIVM in lumbar spine

1- Extension

2- Flexion





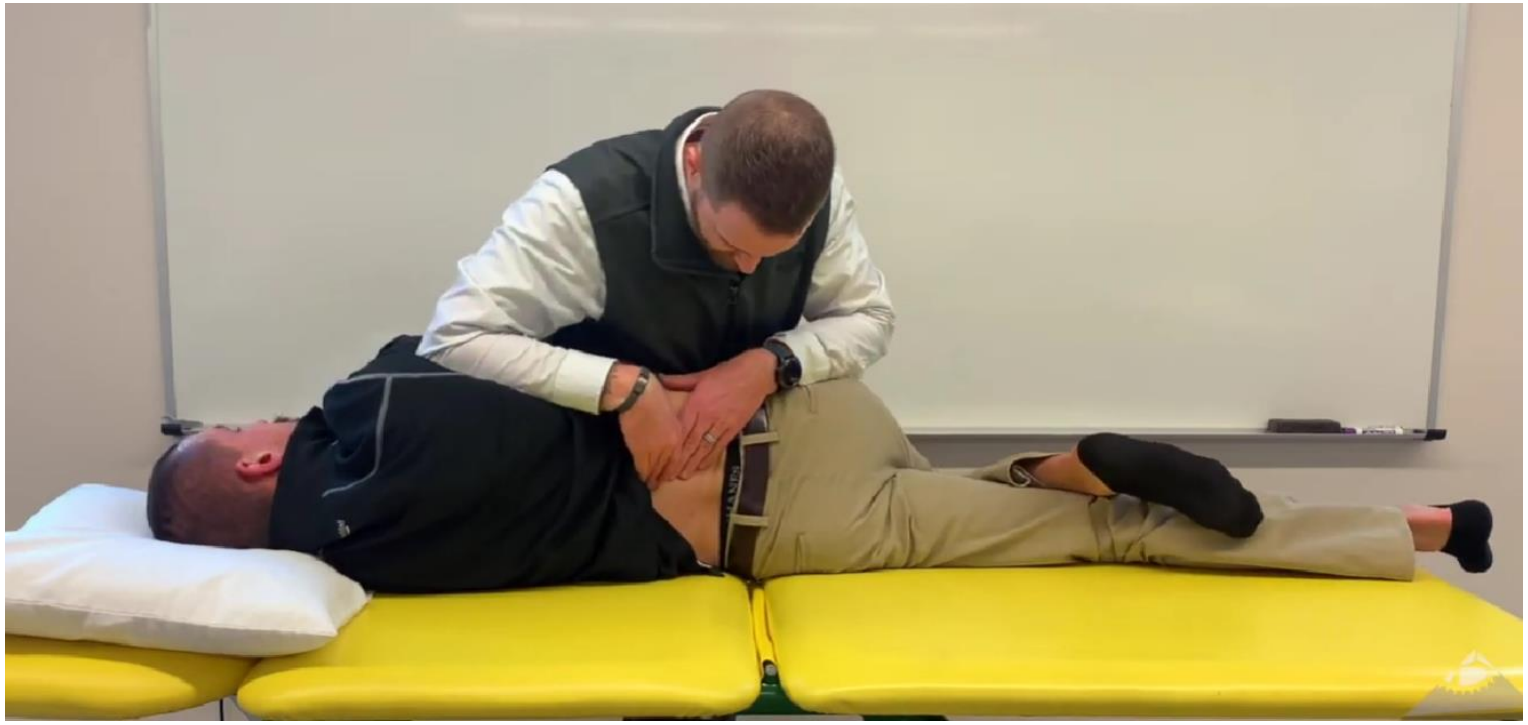


PPIVMS Lumbar side bending





PPIVMs in Rotation





During PPIVMS assessment

- **What you are looking for during PPIVM assessment**
- During PPIVM assessment you guide the patient's spine through specific physiological movements such as flexion, extension, rotation, or side-flexion while palpating the spinous or transverse processes.
- You are looking for the amount of movement available at each segment, the smoothness of the movement, and whether any levels feel restricted or excessively loose.
- You are also checking whether symptoms are reproduced during the motion.
- This helps you identify segments that are hypomobile, hypermobile, painful, or dysfunctional during normal physiological movement.

During PPIVMS Treatment

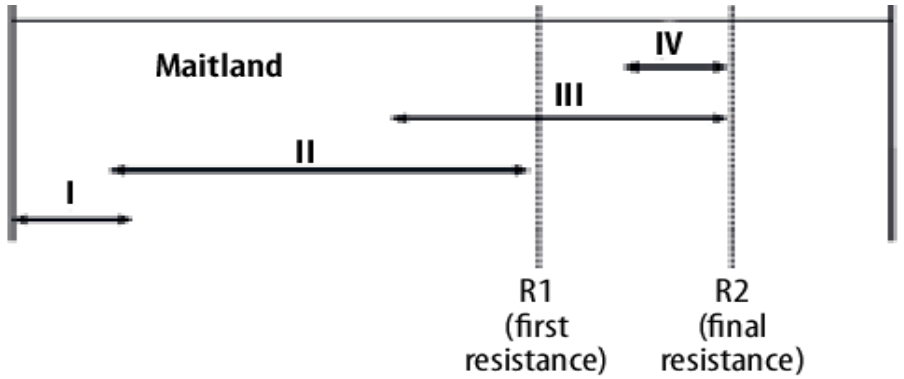
- **What you are looking for during PPIVM treatment**
- During PPIVM treatment you apply repeated passive physiological movements to the restricted segment to improve its mobility and reduce symptoms.
- You look for increasing range, smoother motion, reduced resistance, and less symptom reproduction as the sets progress.
- PPIVMs are often used when the desired functional movement is limited, so treatment aims to restore the specific physiological movement rather than only accessory glide.
- Improvement is confirmed through repeated reassessment of active movement after each set.

1. Maitland's Mobilization

Mobilizations involve repetitive, passive movements applied to spinal joints at varying speeds and amplitudes.

They are graded according to the **Maitland Concept**:

- **Grade I:** A small-amplitude movement near the starting position of the range
- **Grade II:** A large-amplitude of movement that carries well into the range. It can occupy any part of the range that is free of any stiffness or muscle spasm that does move into stiffness or muscle spasm
- **Grade III:** A large-amplitude movement, but one that does move into stiffness or muscle spasm
- **Grade IV:** A small-amplitude movement stretching into stiffness or muscle spasm



During PAIVM Assessment

- During PAIVM assessment we are palpating the intervertebral spinal mobility (between each two vertebrae) when applying a posterior–anterior (PA) glide.
- Assessing whether the segment feels normal, stiff, hypermobile, painful, or different compared with the levels above and below.
- Note whether the pain reproduced matches the patient's symptoms, including location, depth, and behaviour.
- Assessing the quality of the end-feel, the resistance felt under the therapists hands, and any protective muscle guarding.
- Each PA pressure helps determine whether the joint is contributing to the patient's symptoms and whether it needs treatment.

During PAIVM treatment

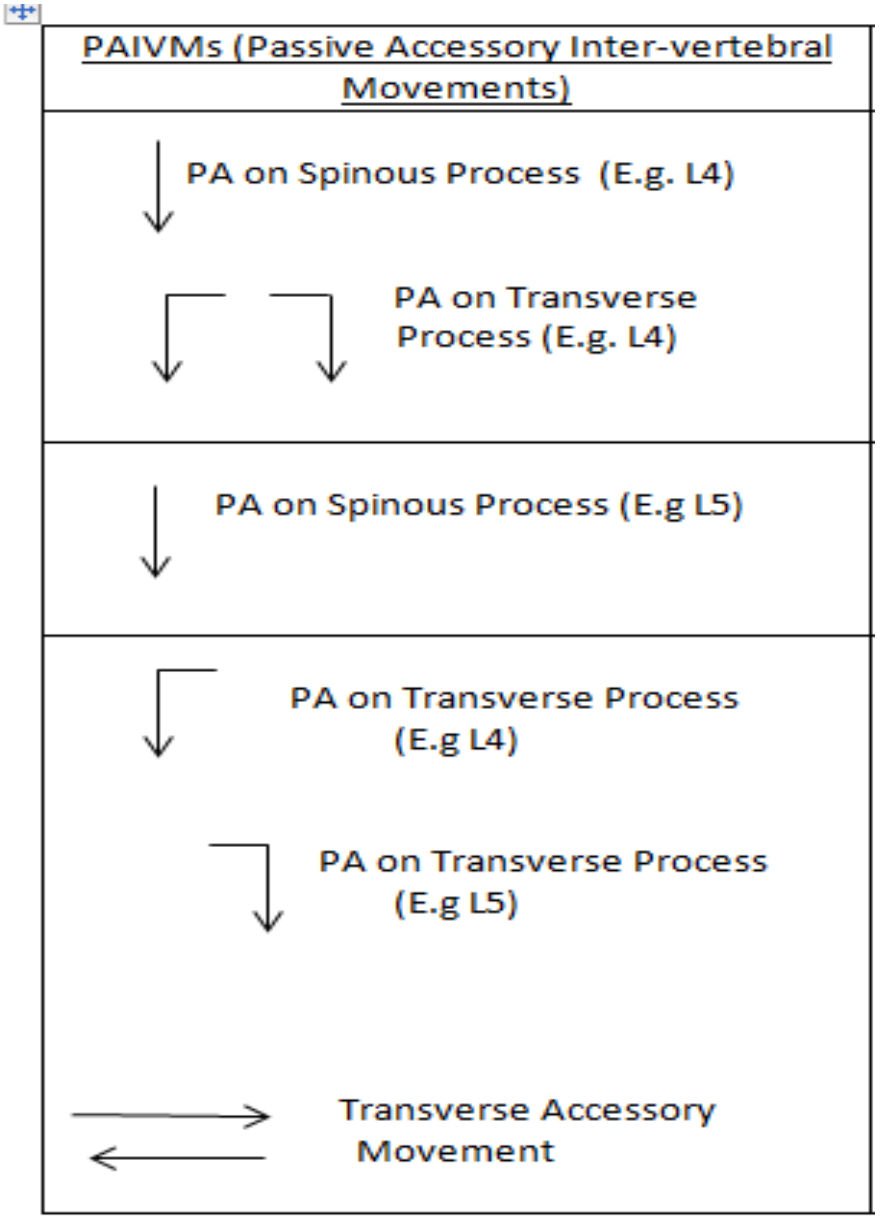
PAIVMs are defined as producing movements in directions that can not be produced actively in isolation but nonetheless need to be available for active movements to occur.

In the periphery, these movements are generally glides produced by a linear force and result in translational rather than rotational movement.

In the spine, PAIVMs are produced by applying a translational force, but unlike in the periphery, the movements that are produced are not purely translational.

Directions of PAIVMS

- PA: Posteroanterior
- AP: Anteroposterior



PAIVMs Lumbar PA Central and Lateral





PAIVMs Lumbar

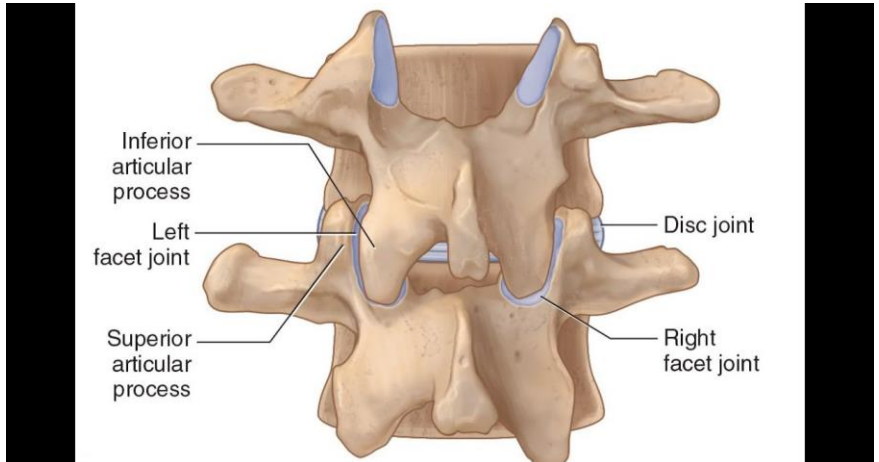
A, PA pressure on spinous process in neutral prone position.

B, Lateral pressure (Transverse) on the spinous process in neutral prone position.

C, PA pressure on the zygoapophyseal joint in neutral position.

D, PA pressure on a spinous process in flexion prone position.

E, Lateral pressure on the spinous process in extension.













Assessment Findings

- **Symptom reproduction or reduction**
  - Location of symptoms?
  - Where in the range?

- **Range of movement**
  - Restricted by pain (pain limited) or stiffness (hypomobility)?
  - Increased (hypermobility)?

- **Quality of movement and feel in movement**
  - Remember normal movement is smooth!

- Affect on functional marker / movement



Dosage of Treatment

- The typical starting point is two to three sets of thirty to sixty seconds.
- Oscillations are applied at a rate of 2 per second prescribe sets of time, as opposed to sets of oscillation number
- Low irritability: 4 sets of 30 seconds with 10 seconds rest
- High irritability: 12 sets of 10 seconds with 30 seconds rest

PAIVMs (Passive Accessory Inter-vertebral Movements)	Comments:
PA on Spinous Process (E.g. L4) PA on Transverse Process (E.g. L4)	All assist flexion at the lumbar spine
PA on Spinous Process (E.g L5)	Assists Extension at the lumbar spine (E.g, PA on L5 causes L4 to glide AP)
PA on Transverse Process (E.g L4) PA on Transverse Process (E.g L5)	All assist rotation to the right. The same concept can be concluded for rotation to the left.
Transverse Accessory Movement	Assists rotation of the lumbar spine


Questions and Break

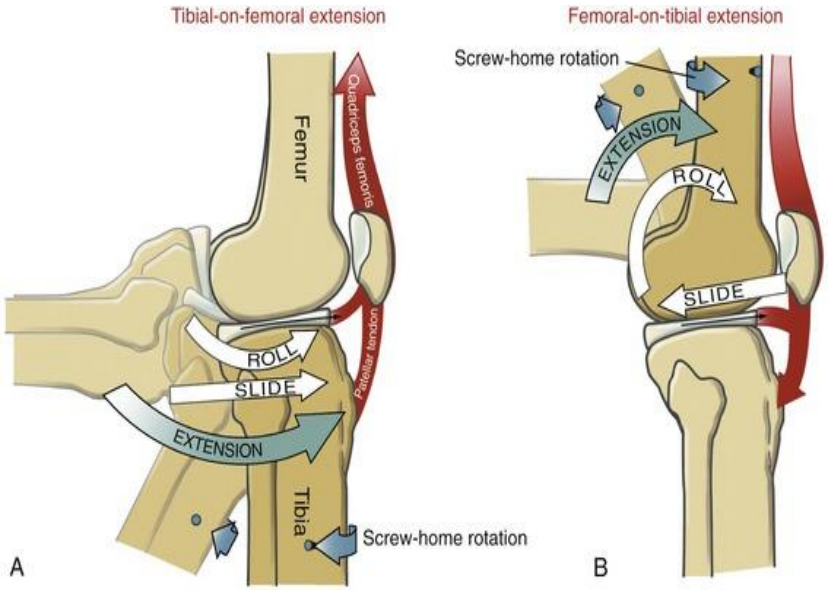


Mulligan Technique

Some of the pics in this section are taken from MR. Asyhara Naela Arifin resource online.

Concept

- Mulligan proposed that injuries or sprains might result in a minor "positional fault" to a joint causing restrictions in physiological movement.
- The techniques have been developed to overcome joint `tracking' problems or `positional faults', i.e. joints with subtle biomechanical changes.
- Normal joints have been designed in such a way that the shape of the articular surfaces, the thickness of the cartilage, the orientation of the fibres of ligaments and capsule, the direction of pull of muscles and tendons, facilitate free but controlled movement while simultaneously minimizing the compressive forces generated by that movement
- Normal proprioceptive feedback maintains this balance. Alteration in any or all of the above factors would alter the joint position or tracking during movement and would provoke symptoms of pain, stiffness or weakness in the patient. It is common sense then that a therapist would attempt to re-align the joint surfaces in the least provocative way



Principles of Treatment

- A fundamental rule of Mulligan spinal work is that the therapist applies the accessory glide while the patient performs the symptomatic movement at the same time.
- This combined approach ensures that the glide directly influences the joint's movement pathway.
- The technique is only considered valid if it produces an immediate, observable improvement.
- If a glide does not change symptoms on the first or second attempt, it is abandoned and another direction or level is tried.
- Mulligan cautions clinicians never to persist with a non-effective glide.
- Therapists must ensure that the applied force is sustained throughout the patient's entire movement arc.

- While applying "MWMS" as an assessment, the therapist should look for **PILL response** to use the same as a Treatment .

  P- Pain free.

  I- Instant result.

  LL- Long Lasting.

The second principle is **CROCKS**



C- Contra-indications (No PILL response is a contraindication)



R - Repetitions (Only three reps on the day one)



O- Over pressure



C- Communications



K - Knowledge (of treatment planes and pathologies)



S- Sustain the mobilization throughout the movement.

Techniques

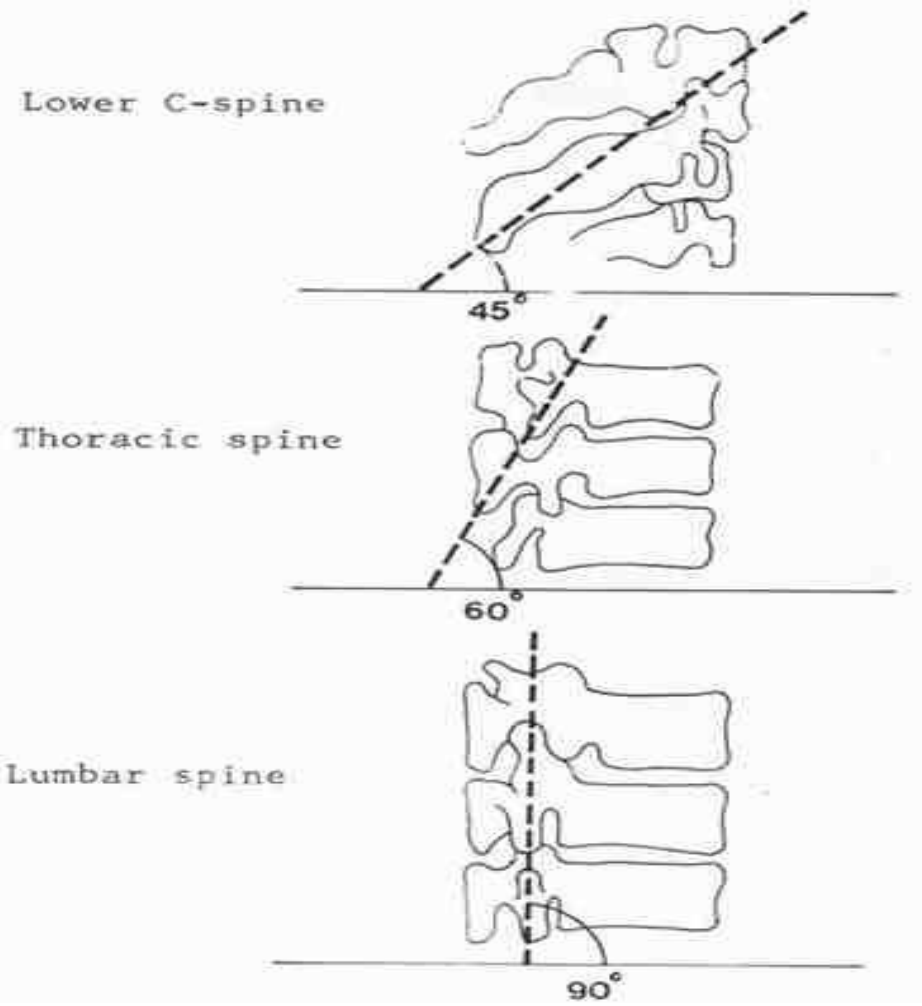
NAGs Natural Apophyseal Glides

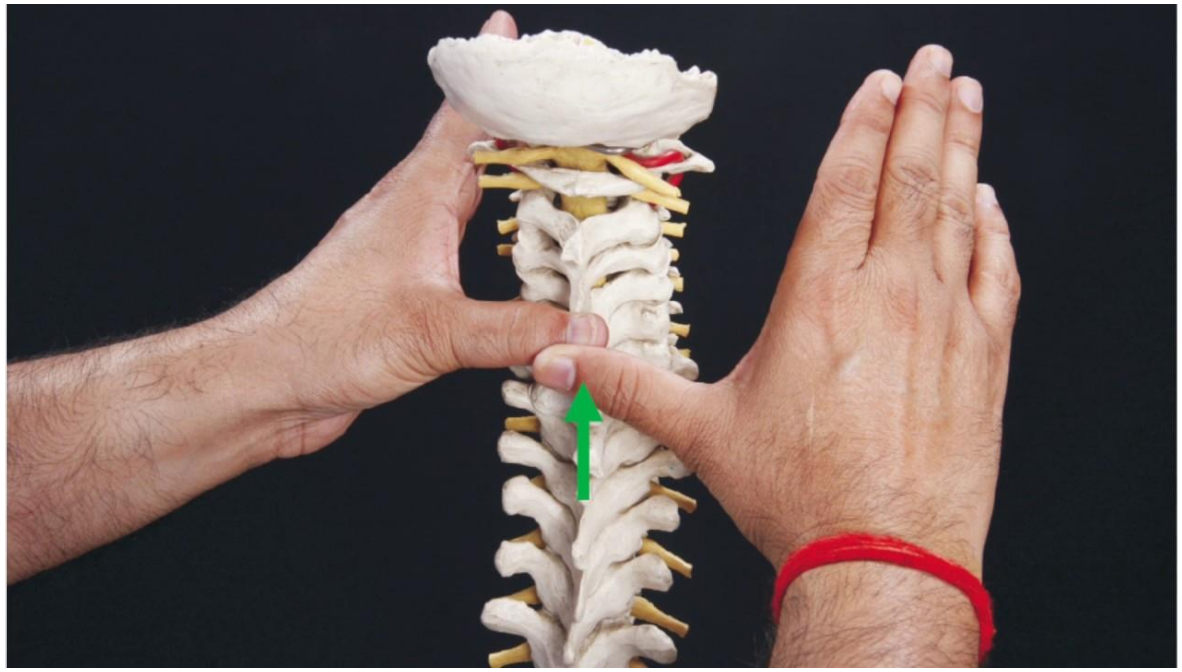
SNAGs Sustained Natural Apophyseal Glides

MWM Mulligans With Movement

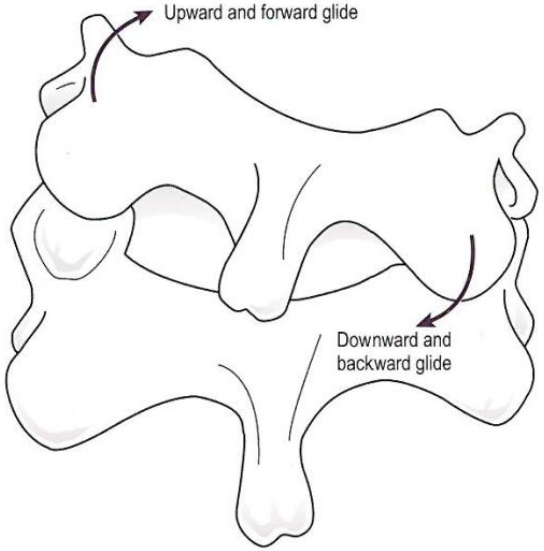
Technique Name	Spinal Region	Description of Technique	Primary Clinical Purpose
SNAGs	Cervical, thoracic, lumbar	The therapist applies a sustained facet glide in the plane of the joint while the patient performs the painful movement. The glide must be painless and must immediately improve the movement.	Used to improve painful end-range movements such as rotation in the cervical spine or flexion and extension in the lumbar spine. Produces rapid change when a positional fault is present.
Self-SNAGs	Cervical, thoracic, lumbar	The patient uses a strap or belt to recreate the therapist's glide and performs the movement independently. This is usually given after a successful manual SNAG to maintain gains.	Reinforces corrected joint mechanics, improves patient independence, and extends treatment effects between sessions.
SMWLM (Spinal Mobilisation With Limb Movement)	Lumbar and thoracic	The therapist applies a sustained glide to the targeted spinal segment while the patient performs a limb movement that previously aggravated symptoms, such as straight-leg raise or hip flexion.	Used for cases where limb movement provokes spinal or referred pain, helping restore pain free neurodynamic or mechanical mobility.
MWM for Sacroiliac Joint (though technically not spine, commonly included)	Lumbopelvic region	The therapist applies a force to reposition the innominate or sacrum while the patient performs a painful functional movement such as forward bending or sit-to-stand.	Helps correct perceived positional faults in the SIJ and restore pain free functional movement in the lumbopelvic area.
Sustained Glide with Traction/Compression Variations	Cervical and lumbar	A SNAG is combined with gentle traction or compression to find the most pain free mechanical environment. The therapist adjusts the position until symptoms reduce.	Provides symptom modification when standard SNAGs alone are not sufficient. Useful for radicular symptoms or loading sensitivity.

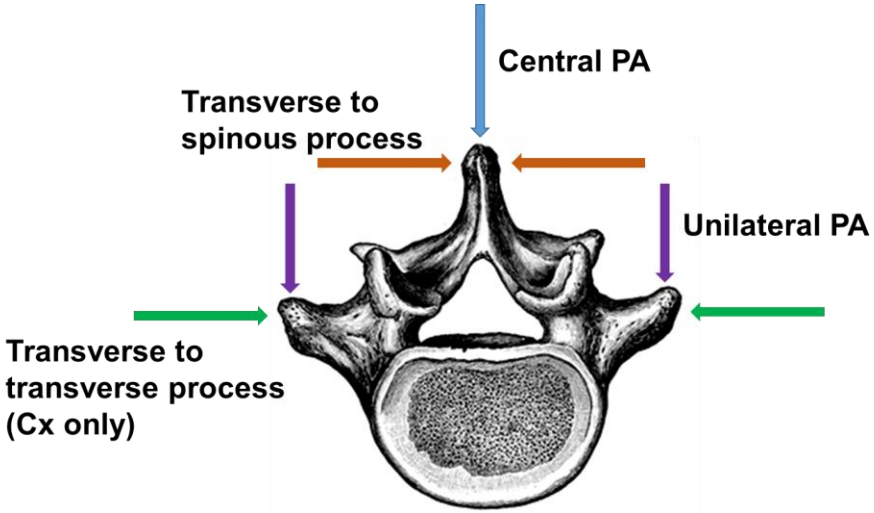
Direction of mobilization application

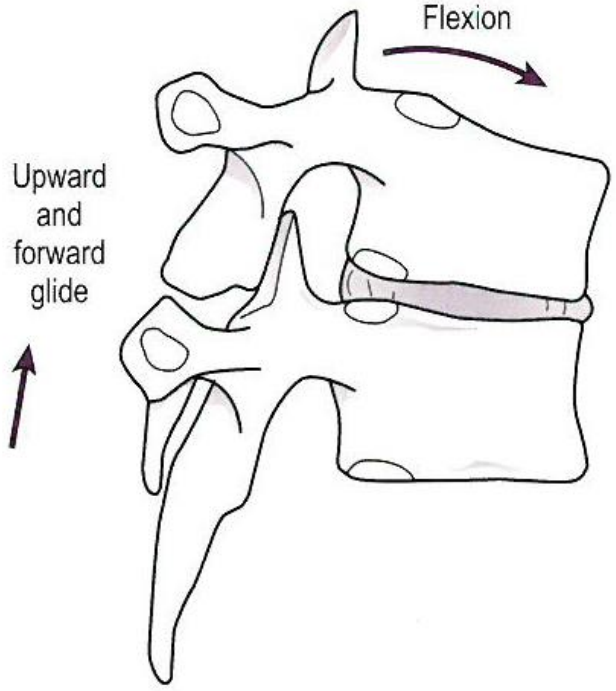


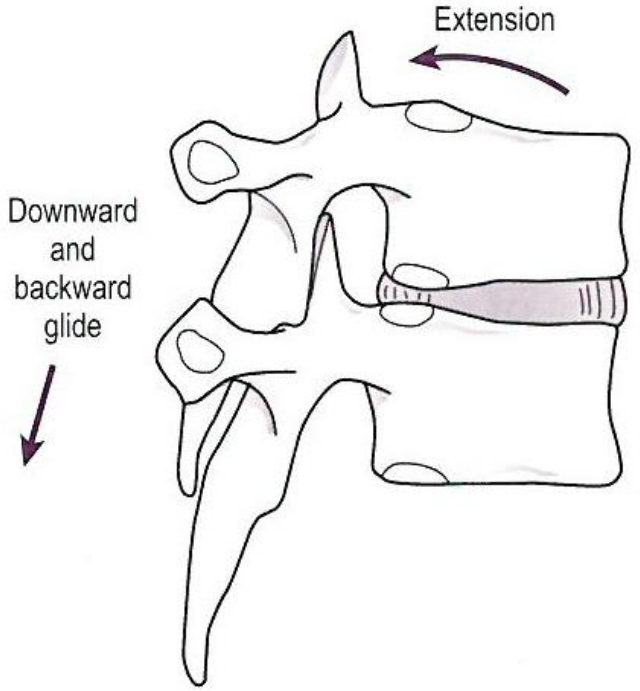


Direction of mobilization





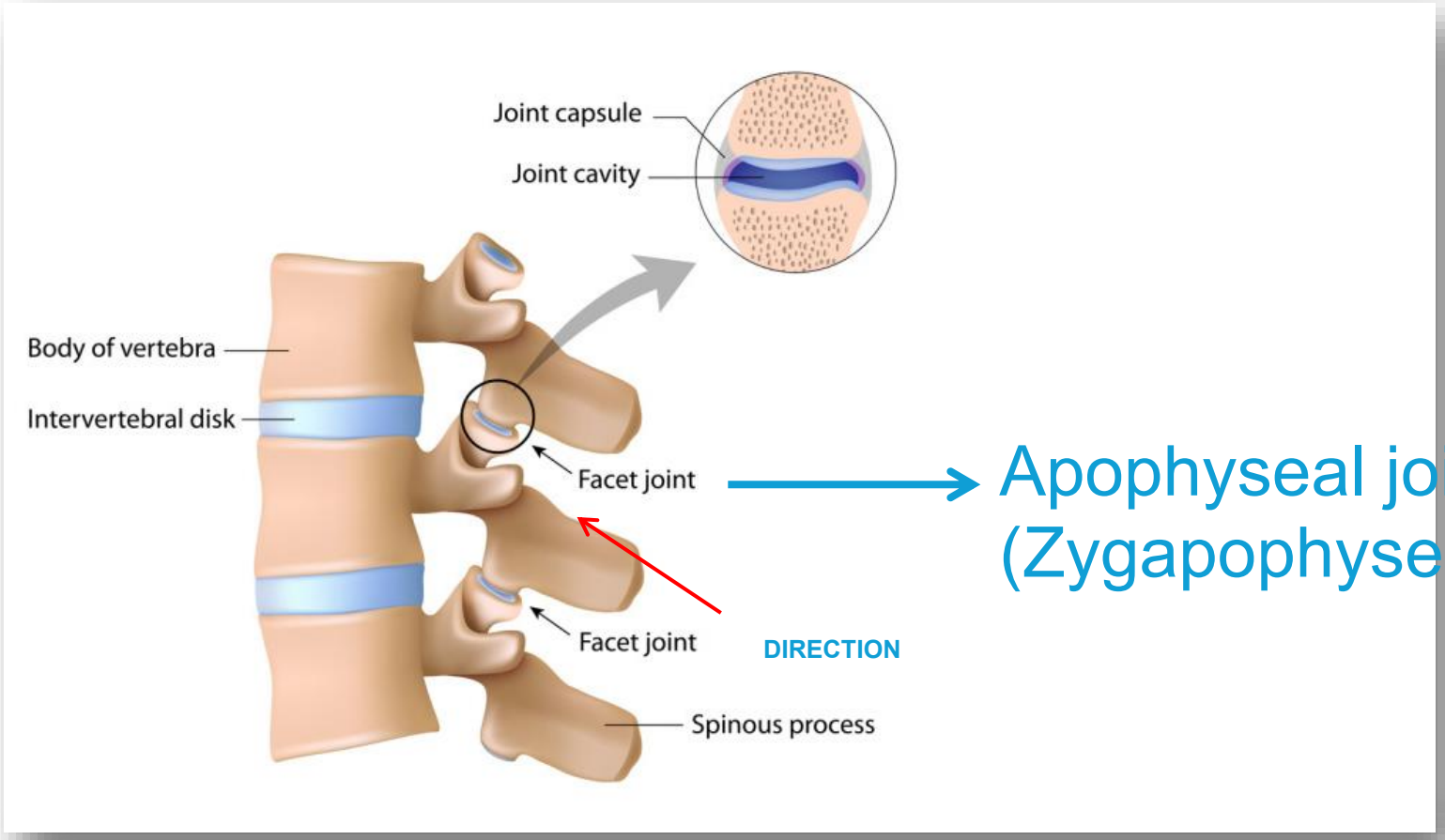




Does

- Repetitive movements leads to a re-learning of the ‘correct’ position
- According to Mulligan ‘several’ repeats
- This has been documented as 6-10reps with 2-3 sets. NOT BLACK AND WHITE

Techniques



Apophyseal joint (Zygapophyseal)

1. Lumbar SNAGS
2. Lumbar Flexion
    1. Lion - NWB
    2. Sitting - PWB
    3. Standing - FWB
    4. Self-SNAG with Belt
3. Lumbar Extension SNAG
    1. Lion – NWB
    2. Prone- PWB
    3. Sitting - PWB
    4. Standing - FWB
    5. Self-SNAG with belt
    6. SNAGS with the Lion Exercises Unilateral L5/S1

Spinal Mobilization with Limb Movement (SMWLMs)

Here a transverse pressure is applied to the side of the relevant spinous process as the patient concurrently moves the limb through the previously restricted range of movement.

The assumption here is that the restriction of movement is of spinal origin of course.

This does not necessarily imply neural compromise since spinal movement must occur when a limb moves beyond a certain point.

Thus the technique addresses a spinal structural/ mechanical restriction, but this may have neural implications too.

Lumbar SNAG

Start Position

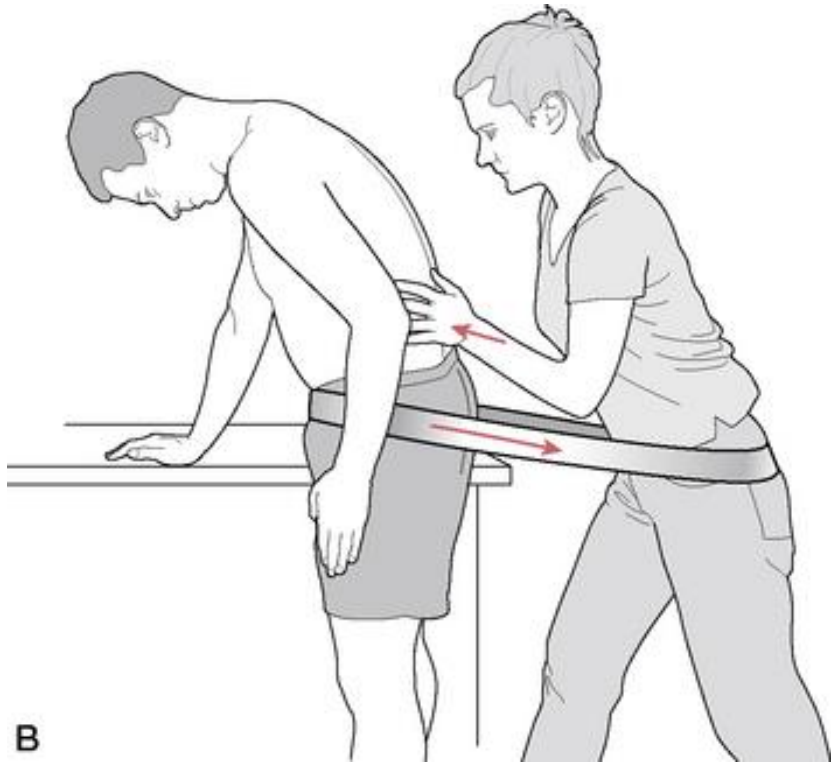


Thumb contact



A: SNAGs for lumbar flexion restriction, patient seated.
B: SNAGs for lumbar flexion restriction, patient standing
C: SNAGS for Lumbar Flexion restriction, patient quadruped (Lion)







SNAGs - Lx

- ACCESSORY + PHYSIOLOGICAL MOVEMENT
- Pt seated (may need feet on chair) (can be standing)
- Therapist stands behind
- Pillow/towel on their tummy
- Belt/strap around their hips and your hips
- Palpate Lx to locate relevant vertebra
- Administer glide
- Ask patient to actively move to the problem movement (exhale when doing flexion)
- Up to 6 reps (maybe only 3 on first Rx)
- Patient can apply overpressure at EOR if pain free

e.g. PA + extension/flexion/side flexion/rotation

Unilateral/transverse + rotation:- you do side lunge





SNAGS in Extension





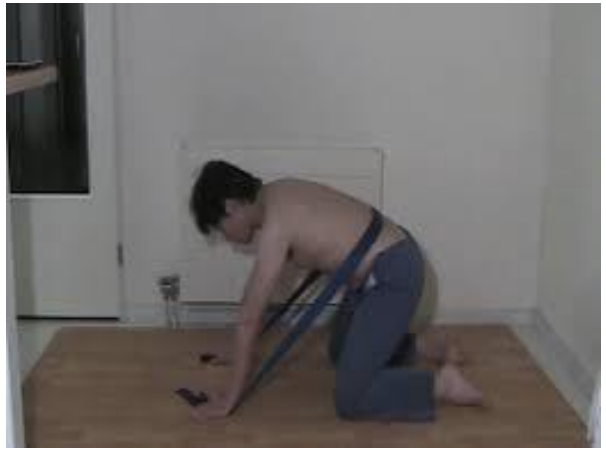




SNAGS in Rotation and Side-bending







Self SNAG with a belt









Soft Tissue Techniques (Myofascial Release & Trigger Point Therapy)

- **Myofascial release:** Sustained pressure applied to restricted fascia (e.g., paraspinal muscles).
- **Trigger point therapy:** Ischemic compression applied to hyperirritable muscle knots (e.g., upper trapezius for neck pain).
- Ajimsha et al. (2015) found **myofascial release effective in reducing pain and disability in chronic LBP**.
- Travell & Simons (2018) emphasized **trigger point therapy for muscle pain management**.

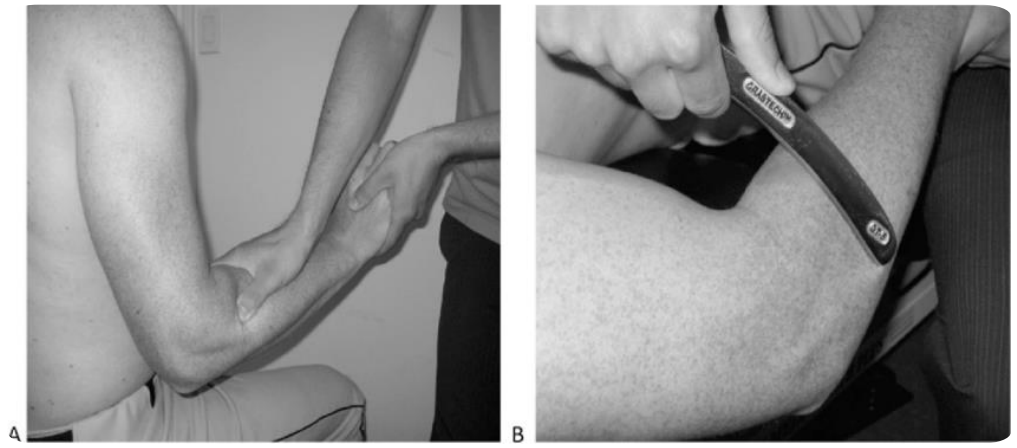
Myofascial Release



- Myofascial release (MFR) therapy involves specifically guided low load, long duration mechanical forces to manipulate the myofascial complex, intended to restore optimal length, decrease pain, and improve function. MFR utilises the manual traction and prolonged stretching of the fascia and muscle to break down the adhesions, thus helps to decrease the pain and increase flexibility and thereby increase ROM.

- It has been hypothesized that fascial restrictions in one part of the body cause undue tension in other parts of the body, due to fascial continuity; in turn, this may create stress on any structures that are enveloped, divided, or supported by fascia (Schleip et al, 2003).

Types of MFR:



- Active Release Techniques (MTrPs Therapy)
- Instrument Assisted Soft Tissue Mobilization
- Direct and indirect Myofascial Release
- Positional Release Technique

Assessment of MFR:

1- tissue rebound

2- Tissue glide

3- traction and compression

MFR Techniques

- Cross hand technique
- Longitudinal Plan Release
- Compression Release
- Transverse plan release



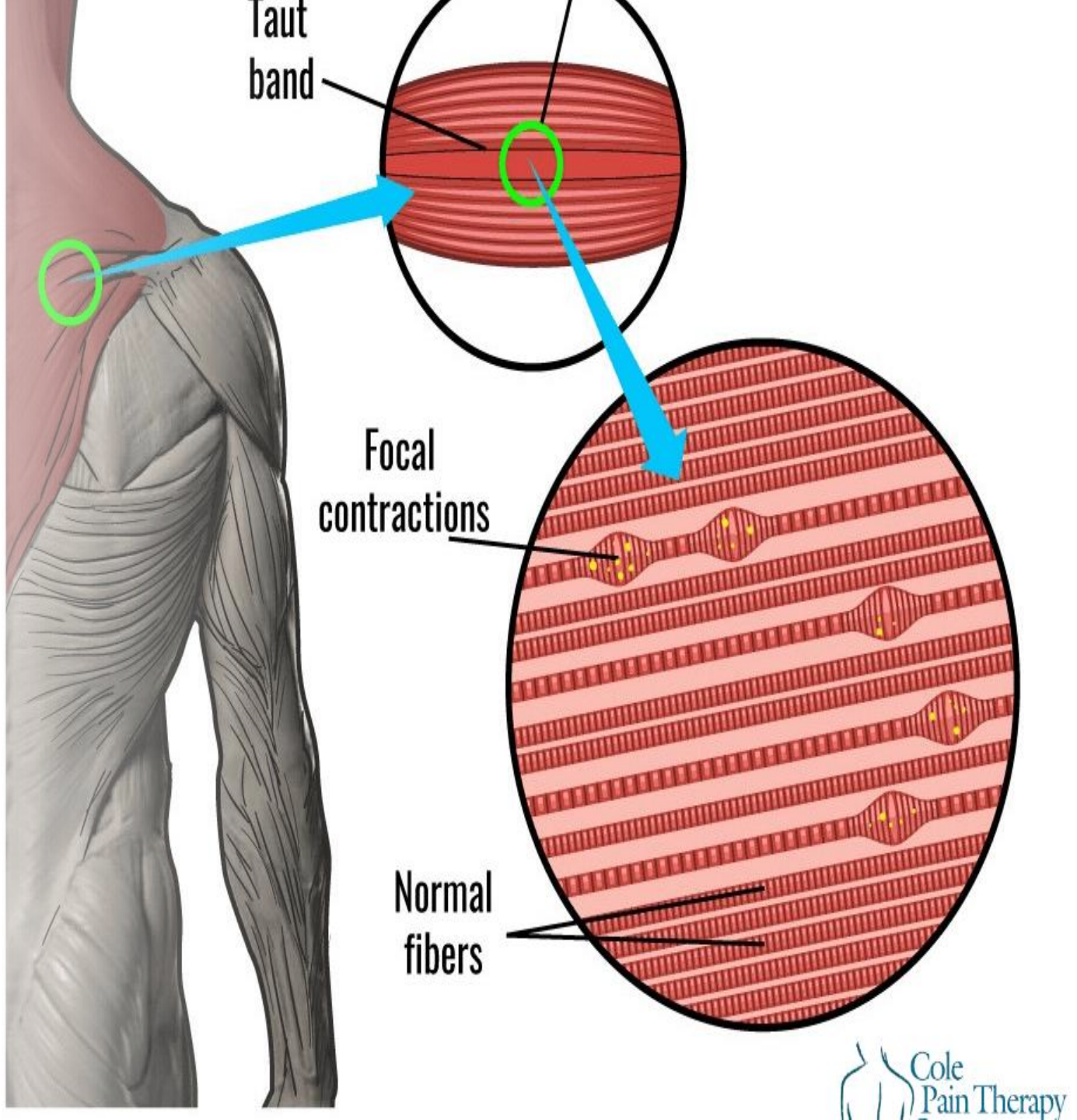






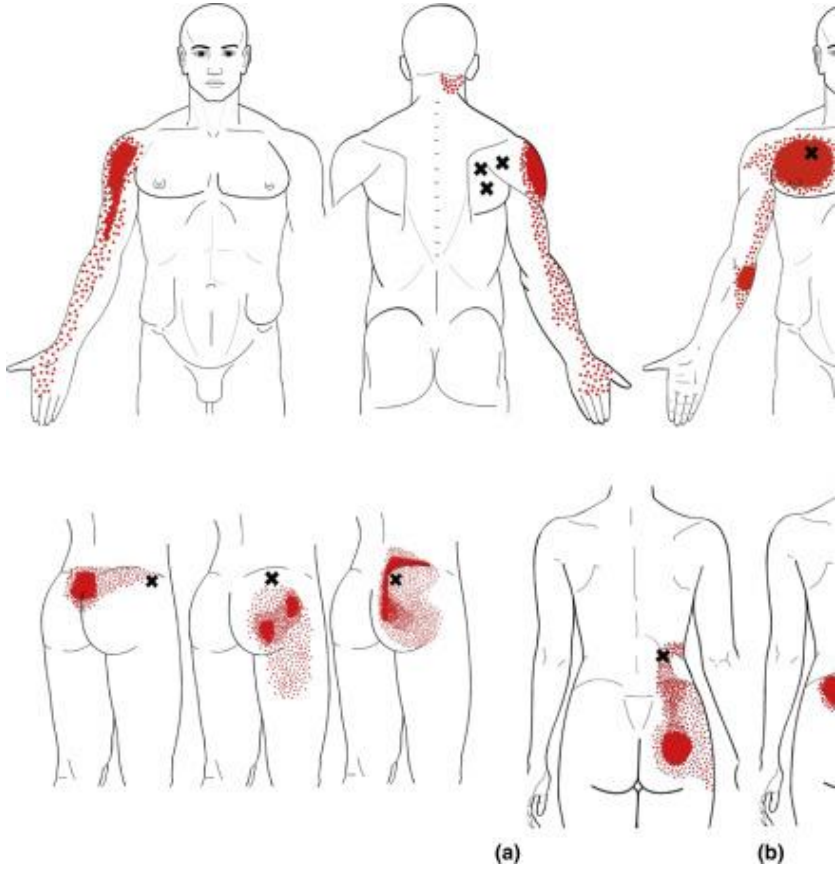
Myofascial Trigger Points (MTrPs)

- Simon's original definition of MPS remains the best; he defined it as a “complex of sensory, motor, and autonomic symptoms that are caused by myofascial trigger points”.
- TrPs are defined as tender or hyperirritable areas in muscles and/or their fascia that are localized in taut, palpable bands, which mediate a local twitch response (a small, quick contraction of muscle fibers that occurs with radiation of pain) under snapping palpation

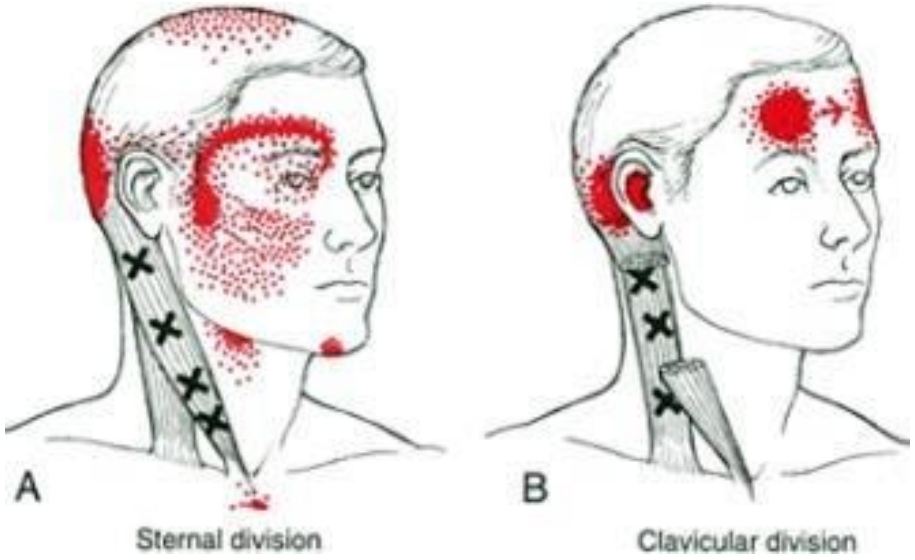


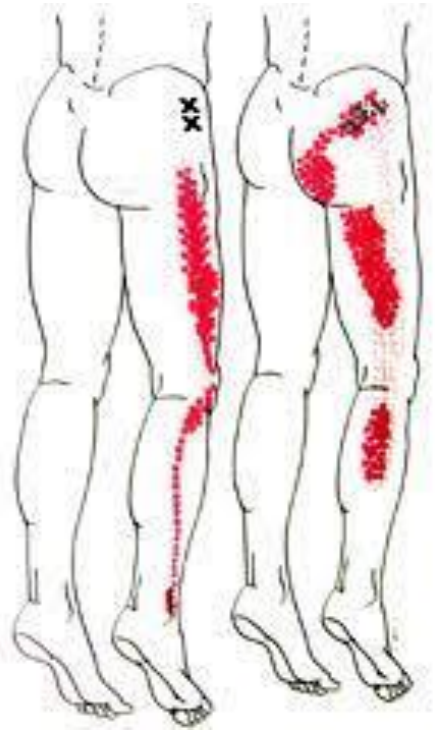
Diagnostic Criteria of MTrPs

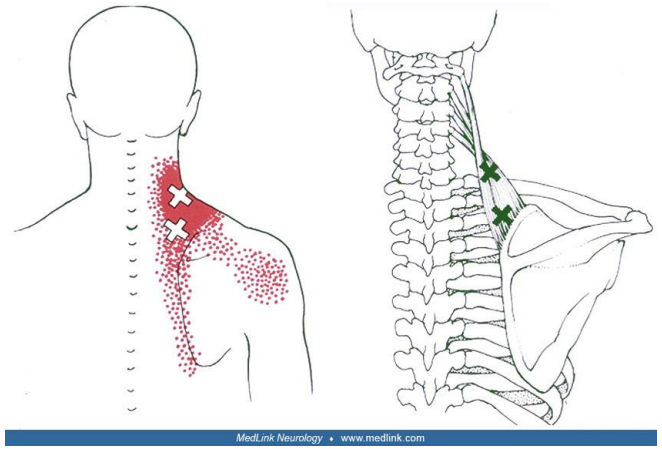
- MPS patients, the distribution of pain is regional, not diffuse
- Palpable taut band of muscle-containing hyperirritable nodules.
- jump sign, and Local Twitch Response
- Referred pain may also be induced by compression of either active or latent TrPs .
- Other potential associated clinical observations in MPS patients include weakness without atrophy, decreased range of motion, and a local twitch response.

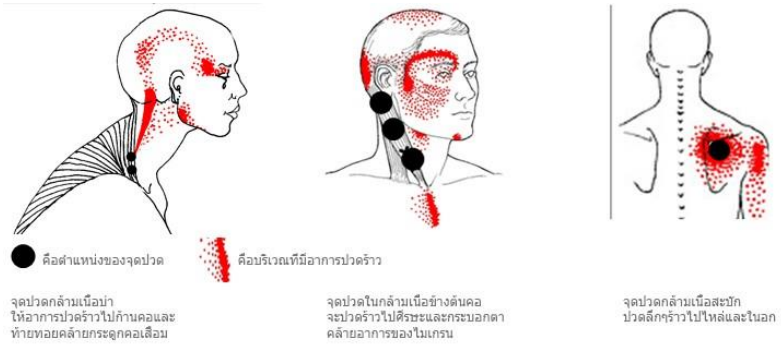


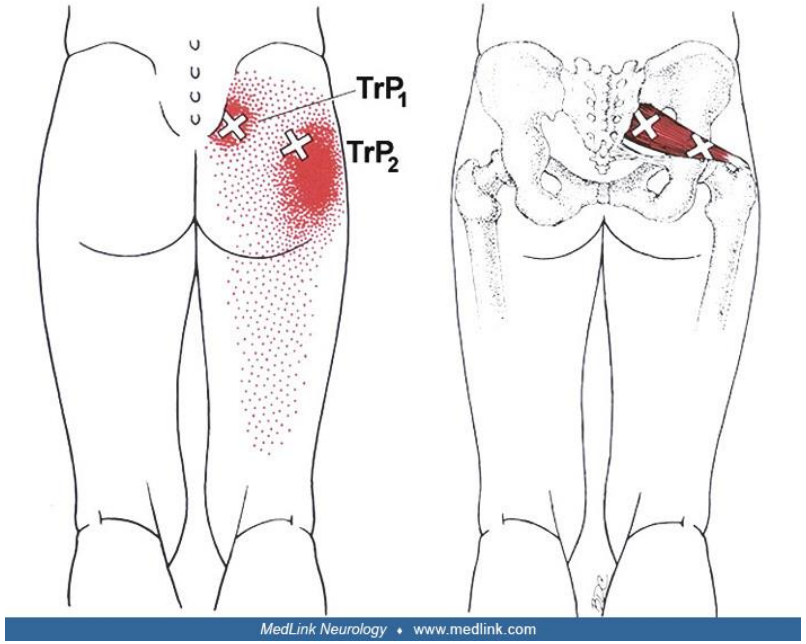
MTrPs


A
Sternal division
B
Clavicular division




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คือตำแหน่งของจุดปวด
คือบริเวณที่มีอาการปวดร้าว
จุดปวดกล้ามเนื้อบ่า
ให้อาการปวดร้าวไปก้านคอและ
ท้ายทอยคล้ายกระดูกคอเสื่อม
จุดปวดในกล้ามเนื้อข้างต้นคอ
จะปวดร้าวไปศีรษะและกระบอกตา
คล้ายอาการของไมเกรน
จุดปวดกล้ามเนื้อสะบัก
ปวดลึกๆร้าวไปไหล่และในอก


TrP1
TrP2
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Key Differences Between Myofascial Pain Syndrome and Fibromyalgia

Feature	Myofascial Pain Syndrome	Fibromyalgia
Trigger/Tender points	Few, localized trigger points	Multiple, generalized tender points
Musculoskeletal pain	Localized	Generalized
Pain in a different spot than its source	More frequent	Less frequent
Fatigue	Less frequent	More frequent
Poor sleep	Less frequent	More frequent
Burning, prickling or tingling sensation	Less frequent	More frequent
Headache	Less frequent	More frequent
Irritable bowel	Less frequent	More frequent
Sensation of swelling	Less frequent	More frequent

Hans SC, Harrison P. MPS and TP management. *Reg Anesth*. 1999;22(1):89-101.
Adapted by Chandola HC, Chakraborty A. Fibromyalgia and myofascial pain

Treatments

- ✓Intermittenet Ischemic Compression
- ✓Pressure Release
- ✓ integrated neuromuscular inhibition technique
- ~~Dry Needling~~
- ~~Injections Therapy~~







